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Nuclear and Mitochondrial Interaction Involving *mt-Nd2* Leads to Increased Mitochondrial Reactive Oxygen Species Production*[S]

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NADH dehydrogenase subunit 2, encoded by the mtDNA, has been associated with resistance to autoimmune type I diabetes (T1D) in a case control study. Recently, we confirmed a role for the mouse ortholog of the protective allele (*mt-Nd2*^{a}) in resistance to T1D using genetic analysis of outcrosses between T1D-resistant ALR and T1D-susceptible NOD mice. We sought to determine the mechanism of disease protection by elucidating whether *mt-Nd2*^{a} affects basal mitochondrial function or mitochondrial function in the presence of oxidative stress. Two lines of reciprocal conplastic mouse strains were generated: one with ALR nuclear DNA and NOD mtDNA (ALR.mt^{NOD}) and the reciprocal with NOD nuclear DNA and ALR mtDNA (NOD.mt^{ALR}). Basal mitochondrial respiration, transmembrane potential, and electron transport system enzymatic activities showed no difference among the strains. However, ALR.mt^{NOD} mitochondria supported by either complex I or complex II substrates produced significantly more reactive oxygen species when compared with both parental strains, NOD.mt^{ALR} or C57BL/6 controls. Nitric oxide inhibited respiration to a similar extent for mitochondria from the five strains due to competitive antagonism with molecular oxygen at complex IV. Superoxide and hydrogen peroxide generated by xanthine oxidase did not significantly decrease complex I function. The protein nitrating agents peroxynitrite or nitrogen dioxide radicals significantly decreased complex I function but with no significant difference among the five strains. In summary, *mt-Nd2*^{a} does not confer elevated resistance to oxidative stress; however, it plays a critical role in the control of the mitochondrial reactive oxygen species production.

Single nucleotide polymorphisms in the mtDNA have been associated with degenerative diseases and various cancers. Yet sequence changes may also result in resistance to disease. Indeed, a cytosine to adenine transversion (C5178A) resulting in a leucine to methionine substitution in the human NADH dehydrogenase subunit 2 gene (*mt-ND2*) encoded in the mtDNA has been associated with increased longevity (1, 2) as well as reductions in atherosclerosis (3), blood pressure (4), myocardial infarction (5), and T1D[3] incidence (6). The adenine-containing allele, *mt-ND2*^{a} (adenine containing NADH dehydrogenase subunit 2 allele), was also associated with reduced islet autoimmunity as represented by significantly lower titers of autoantibodies against glutamic acid decarboxylase, insulin, and protein-tyrosine phosphatase, receptor type N (Ptprn or IA-2) (6).

Insulin synthetic and secretory capacities of pancreatic β cells are highly dependent upon communication between the nucleus and the mitochondria (7) and are reliant upon mitochondrial ATP generation (8). Although mitochondria are critical for the life and function of β cells, there is strong evidence that mitochondria play a central role in apoptotic death of the β cell during autoimmune T1D (9–16). Therefore, sequence variation in the mtDNA may have profound effects on the β cell, both in life and death.

The function of ND2 is not clearly understood. Studies of *Escherichia coli* complex I have established that the bacterial ortholog of ND2, NuoN, folds into one of the many α-helices in the membrane arm of complex I (17, 18). The current understanding of the mechanism of complex I suggests that ND2 is involved in proton translocation across the inner mitochondrial membrane (17, 19–21), and its protein sequence shows similarity to an antiporter that is critical in pH regulation (20). ND2 is a hydrophobic subunit of complex I that has been conserved through the course of evolution from bacteria to mice to humans (22).

Recently, our group has confirmed a role for NADH dehydrogenase subunit 2 gene (*mt-Nd2*) in resistance to diabetes using crosses of the T1D-prone non-obese diabetic (NOD) mouse strain to the alloxan-resistant (ALR) mouse strain (23). Reciprocal backcross populations were generated with identical nuclear DNA but either ALR or NOD mtDNA. Spontaneous

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[3] The abbreviations used are: T1D, type 1 diabetes; ALR, alloxan-resistant mouse strain; NOD, non-obese diabetic mouse strain; ROS, reactive oxygen species; B6, C57BL/6J; NO, nitric oxide; CS, conplastic mouse strain; ETS, electron transport system; FCCP, carbonyl cyanide *p*-trifluoromethoxyphenylhydrazone; ETFDH, electron transfer flavoprotein dehydrogenase; ALS, alloxan-susceptible.

T1D incidence was 4-fold lower in the backcross population with ALR mtDNA than in the backcross population with NOD mtDNA (23). In both man and mouse the "A"-containing allele, *mt-ND2^a* or *mt-Nd2^a*, respectively, resulted in a leucine to methionine amino acid substitution and contributed to T1D resistance.

ALR mice, which encode *mt-Nd2^a*, resist both chemically induced and autoimmune T1D. Along with *mt-Nd2^a*, loci contributing to T1D resistance have been mapped to chromosomes 3, 8, and 17 (24). ALR mice were bred specifically for their resistance to the diabetogen alloxan, which selectively destroys β cells through the generation of hydroxyl radicals (25). Biochemical analyses have revealed that in comparison with the NOD or co-selected alloxan-susceptible (ALS) strain, ALR mice contain elevated reduced to oxidized GSH in the liver and islets, as well as elevated activity of superoxide dismutase in the liver and islets, and elevated activity of glutathione peroxidase and glutathione reductase in islets (26, 27). The destruction of β cells during T1D pathogenesis has been characterized by increased reactive oxygen species (ROS) production and attenuated antioxidant defenses, including decreased levels of GSH and gradually reduced catalase, superoxide dismutase, and thioredoxin transcripts (28–30). Therefore, it has been suggested that increased redox potential in β cells may inhibit T1D by protecting β cells from apoptosis (27, 31, 32). Indeed, GSH, superoxide dismutase, and thioredoxin have been shown to block the actions of signal-regulating kinase, AP-1, and NF-κB, thereby inhibiting apoptosis (33–35).

Mitochondria are both generators (36) and important targets of ROS (37). In this context, it has been proposed that *mt-Nd2^a* imparts upon the mitochondria an increased resistance to oxidative stress (6) as many of the diseases against which *mt-ND2^a* imparts resistance, such as aging and T1D, have been associated with ROS damage. In support of the hypothesis that *mt-Nd2^a* results in elevated resistance to oxidative stress, Turko *et al.* (38) have reported that complex I enzymatic activity of ALR was not affected by free radicals, although the same treatments reduced complex I activity in ALS by ~50%.

The aim of this study was to determine the effect of *mt-Nd2^a* on basal mitochondrial function as well as on mitochondrial function in the presence of oxidative stress to further characterize its role in the protection against T1D. To study the specific role of *mt-Nd2^a*, two lines of reciprocal conplastic mouse strains (CS) were developed, one with ALR nuclear DNA and NOD mtDNA (ALR.mt^{NOD}) and one with NOD nuclear DNA and ALR mtDNA (NOD.mt^{ALR}). By combining *mt-Nd2^a* with NOD nuclear DNA, the effects of this allele could be considered separately from the protective effects of the nuclear genome of ALR. Here we assess basal mitochondrial functions and determine the resistance of mitochondria to free radicals. We find that *mt-Nd2^a* does not confer elevated resistance to oxidative stress. However, *mt-Nd2^a* suppresses mitochondrial ROS production.

EXPERIMENTAL PROCEDURES

Mice—ALR/LtJ, NOD/LtJ, and C57BL/6J (B6) mice were bred and maintained in the animal research facility at the Rangos Research Center, Pittsburgh, PA. Conplastic strains of mice, NOD/LtJ-mt$^{ALR/LtJ}$/Mx (NOD.mt^{ALR}) and ALR/LtJ-mt$^{NOD/LtDvs}$/Mx (ALR.mt^{NOD}), were generated as described below. All mice were bred and maintained in a specific pathogen-free vivarium and allowed free access to food (autoclaved diet NIH-31, 6% fat, PMI, St. Louis, MO) and acidified drinking water. All procedures involving animals were approved by the Children's Hospital of Pittsburgh and were in compliance with "Principles of Laboratory Animal Care" and the current laws of the United States.

Reagents—All reagents were obtained from Sigma unless otherwise noted.

Generation of Reciprocal Conplastic Strains of Mice—Lines of reciprocal CS mice were generated to determine the role of *mt-Nd2^a* in resistance to T1D and for any possible effects on mitochondrial function. Strains with ALR nuclear DNA and NOD mtDNA (ALR.mt^{NOD}) or with NOD nuclear DNA and ALR mtDNA (NOD.mt^{ALR}) were generated as described previously (39) with minor modifications. Because mtDNA is inherited exclusively from the egg, only female breeders with the appropriate mtDNA were employed. The generation of ALR.mt^{NOD} CS utilized an F1 outcross of ALR males to NOD females resulting in F1 progeny with NOD mtDNA. Females of this outcross were then backcrossed to ALR males for 10 generations, allowing for continued inheritance of the NOD mtDNA. Conversely, to generate NOD.mt^{ALR} CS, NOD males were outcrossed to ALR females resulting in F1 progeny with ALR mtDNA. At each generation females were backcrossed to NOD males until the 10th backcross generation and then intercrossed. Single nucleotide polymorphism typing was conducted to determine the *mt-Nd2* allele in the CS as described (23). To preclude nuclear DNA contamination in the CS mice, genotyping was performed by PCR amplification of 94 polymorphic microsatellite primers (Invitrogen) covering all 19 autosomes (24) (supplemental Table).

Liver Mitochondrial Isolation—Livers were removed and homogenized in ice-cold isolation buffer (IB) I (225 mM mannitol, 75 mM sucrose, 10 mM HEPES potassium salt, 0.10% bovine serum albumin, fatty acid-free, and 1 mM EDTA, pH 7.4). The homogenate was centrifuged at 1,300 $\times$ *g* for 10 min. The supernatant was transferred into new tubes, diluted with IB I, and centrifuged at 10,000 $\times$ *g* for 10 min. The supernatant was discarded, and the pellet was resuspended in IB II (225 mM mannitol, 75 mM sucrose, 10 mM HEPES potassium salt, and 100 μM EDTA, pH 7.4) and spun at 10,000 $\times$ *g* for 10 min. The resulting pellet was resuspended in ~100 μl of IB II.

Brain Mitochondrial Isolation—Brains were removed, homogenized in ice-cold 12% Percoll in IB I, layered on top of a gradient of 24 and 42% Percoll, and centrifuged at 27,000 $\times$ *g* for 10 min. The mitochondrial fraction was removed with a syringe, diluted with IB I, and centrifuged at 10,000 $\times$ *g* for 10 min. The supernatant was discarded, and the pellet was resuspended in IB II and spun at 10,000 $\times$ *g* for 5 min. The supernatant was again discarded, and the pellet was resuspended in ~100 μl of IB II. Protein concentration of both liver and brain mitochondria was determined using the BCA protein assay (Pierce).

Mitochondrial Respiration—Mitochondria (1.6 mg/ml) were incubated in media containing 125 mM KCl, 2 mM K_2HPO_4, 5

mM $MgCl_2$, 10 mM HEPES, and 10 μM EGTA, pH 7.1 (IM). To assay respiration through complex I, the assay medium was supplemented with 5 mM L-glutamate and 5 mM L-malate. To assay respiration through complex II, the assay medium was supplemented with 5 mM succinate. State 3 respiration was stimulated by the addition of 1.81 mM ADP. The respiratory control ratio was calculated by dividing state 3 respiration rates by state 4 respiration rates. Mitochondrial respiration was determined using a Clark-type oxygen electrode (Hansatech Instruments Ltd., Norfolk, UK). Assays were performed at 37 °C with constant stirring.

Individual Mitochondrial Electron Transport Chain Complex Enzymatic Activity Assays—For each assay, mitochondria samples were subjected to membrane disruption by freeze-thawing. All assays were run at 30 °C.

The activity of complex I (NADH:ubiquinone oxidoreductase) was determined by monitoring the oxidation of NADH at 340 nm. The assay medium contained potassium phosphate (25 mM, pH 7.2 at 20 °C), 5 mM $MgCl_2$, 2.5 mg/ml bovine serum albumin (fraction V), and 2 mM KCN. A base line was established for 1 min after the addition of 0.13 mM NADH, 65 μM ubiquinone$_1$, and 2 μg/ml antimycin A. The reaction was initiated by the addition of mitochondria (200 μg/ml), and the rate of oxidation of NADH was recorded for 3 min. Rotenone (2 μg/ml) was then added, and the rate of change in absorbance was measured for an additional 3 min. Complex I activity was determined by subtracting the rotenone insensitive activity from the total activity.

The activity of complex II (succinate:ubiquinone oxidoreductace) was determined by monitoring the reduction of 2,6-dichloroindophenolate at 600 nm. The assay medium contained potassium phosphate (25 mM, pH 7.2 at 20 °C), 5 mM $MgCl_2$, and 20 mM sodium succinate. Mitochondria (40 μg/ml) were incubated in the assay medium at 30 °C for 10 min. A base line was recorded for 1 min after the addition of 2 μg/ml antimycin A, 2 μg/ml rotenone, 2 mM KCN, and 50 μM 2,6-dichloroindophenolate. The reaction was initiated by the addition of 65 μM ubiquinone$_1$, and the rate of reduction of 2,6-dichloroindophenolate was recorded for 3 min (40).

The activity of complex III (cytochrome *c* reductase) was determined by monitoring the reduction of ferricytochrome *c* at 550 nm. The assay medium contained potassium phosphate (25 mM, pH 7.2 at 20 °C), 5 mM $MgCl_2$, 2.5 mg/ml bovine serum albumin (fraction V), and 2 mM KCN. KCN was included in the assay media to prevent the reoxidation of the product, ferrocytochrome *c*, by cytochrome *c* oxidase. Nonenzymatic activity was recorded for 1 min after the addition of 15 μM ferricytochrome *c*, 2 μg/ml rotenone, 0.6 mM dodecyl-β-D-maltoside, and 35 μM ubiquionol. Ubuiquinol was prepared by dissolving 8 μg of ubiquinone in 1 ml of ethanol; the solution was adjusted to pH 2 with 6 M HCl. Ubiquinone was reduced using excess sodium borohydride. Ubiquinol was extracted into 2:1 (v/v) diethyl ether/cyclohexane, evaporated under nitrogen gas, dissolved in 1 ml of ethanol, and acidified to pH 2 with 6 M HCl. The complex III activity assay was initiated by the addition of mitochondria (100 μg/ml), and the rate of reduction of ferricytochrome *c* to ferrocytochrome *c* was recorded for 1 min. The activity quickly became nonlinear, and the rate was calculated based on the linear first 10 s. In replicate wells, 2 μg/ml antimycin A was added, and the complex III specific activity was calculated by subtracting the antimycin A insensitive activity from the total activity (40).

Complex IV (cytochrome *c* oxidase) activity was determined by monitoring the oxidation of ferrocytochrome *c* at 550 nm. The assay medium contained 10 mM Tris-HCl and 120 mM KCl, pH 7.0. The nonenzymatic rate was recorded for 1 min after the addition of 2 μg/ml antimycin A, 0.45 mM dodecyl-β-D-maltoside, and mitochondria (2.5 μg/ml). The reaction was initiated by the addition of 11 μM ferrocytochrome *c*, and the rate of oxidation of ferrocytochrome *c* to ferricytochrome *c* was measured for 3 min. The activity quickly became nonlinear, and the rate was calculated based on the linear first 30 s . In replicate wells, 2 μg/ml KCN was added, and the complex IV specific activity was calculated by subtracting the KCN insensitive activity from the total activity. Ferrocytochrome *c* was prepared by reducing ferricytochrome *c* with 0.5 mM dithiothreitol (40).

Mitochondrial Membrane Potential and ROS Production—Membrane potential and free radical production were measured by fluorescence using a Shimadzu RF-5301 spectrofluorimeter (Kyoto, Japan) as described previously (41). All assays were performed with 350 μg of mitochondrial protein suspended in IM plus 5 mM L-glutamate and 5 mM L-malate or 5 mM succinate with constant stirring at 37 °C. ROS production was measured using 2 μM fluorescent Amplex Red dye (Molecular Probes, Eugene, OR) in the presence of 1 unit/ml horseradish peroxidase. The excitation wavelength was 560 nm (slit 1.5 nm), and the emission wavelength was 590 nm (slit 3 nm). Mitochondrial transmembrane potential ($\Delta\psi_m$) was measured using the fluorescence quenching of the cationic dye safranin O (2.5 μM). The excitation wavelength was 495 nm (slit 3 nm), and the emission wavelength was 586 nm (slit 10 nm).

Free Radical Treatment for Mitochondrial Respiration—Mitochondrial respiration was assayed as described above. Approximately 30 s after the addition of ADP, 15 μM nitric oxide (NO) was added using the NO donor diethylamine NONOate (Cayman Chemical, Ann Arbor, MI). Percent oxygen consumption was calculated by comparing chamber oxygen content at the time of NO addition to the chamber oxygen content 1.5 min after the addition of NO. Calculations were based on the percent of control oxygen consumption. To test whether the effect of NO was reversible, bovine hemoglobin (1.32 mg/ml) was added after 5 min of incubation with 15 μM NO.

Free Radical Treatment for Complex I Enzymatic Activity Assay—Mitochondria were treated with either a 3 mM bolus of NO donated by diethylamine NONOate or a 1 μM steady-state level of NO donated by DETA-NONOate (Cayman Chemical, Ann Arbor, MI) for 2 h. Mitochondria were treated with hydrogen peroxide and superoxide generated by 100 μM xanthine and 20 milliunits of xanthine oxidase and incubated for 2 h. Control samples were treated with xanthine without xanthine oxidase. Peroxynitrite or heme peroxidase-dependent reactions were employed to facilitate protein nitration. Mitochondria were treated with 0.25, 0.50, or 1.00 mM peroxynitrite (Upstate Biotechnology, Inc., Lake Placid, NY) and incubated for 1 h. Controls were treated with the equivalent concentration of

TABLE 1

Comparison of the individual enzymatic activities of ETS complexes I–IV for ALR, NOD, B6, NOD.mt^{ALR}, and ALR.mt^{NOD}

Complex I was assayed as the rotenone-sensitive activity following the oxidation of NADH. Complex II was assayed following the reduction of 2,6-dichloroindophenolate (DCIP). Complex III was assayed as the antimycin A-sensitive activity following the reduction of cytochrome *c* (Cyt *c*) over 10 s. Complex IV was assayed as the KCN-sensitive activity following the oxidation of cytochrome *c* over 30 s. For details see "Experimental Procedures." Values are reported as mean ± S.D. For each strain, $n \geq 3$, with mitochondria from each mouse run in triplicate.

Strain	Complex I (nmol of NADH oxidized)	Complex II (nmol of DCIP reduced)	Complex III (nmol of Cyt *c* reduced)	Complex IV (nmol of Cyt *c* oxidized)
	min × mg protein	*min × mg protein*	*min × mg protein*	*min × mg protein*
ALR	35.83 ± 4.61	62.76 ± 9.10	125.94 ± 12.73	282.14 ± 65.58
NOD	36.10 ± 6.24	62.52 ± 13.26	124.70 ± 8.96	256.76 ± 48.14
B6	34.70 ± 6.04	62.54 ± 7.33	121.51 ± 9.25	285.65 ± 34.59
NOD.mt^{ALR}	35.41 ± 3.15	64.13 ± 10.69	126.12 ± 4.62	263.53 ± 18.56
ALR.mt^{NOD}	35.78 ± 4.13	64.45 ± 8.68	128.19 ± 15.09	273.42 ± 37.78

TABLE 2

Comparison of complex I- or complex II-supported respiration for mitochondria from ALR, NOD, B6, NOD.mt^{ALR}, and ALR.mt^{NOD}

Respiration was assayed using a Clark oxygen-type electrode using the complex I substrates glutamate and malate or the complex II substrate succinate. State 3 respiration was stimulated by the addition of 1.81 mM ADP. Respiratory control ratio was calculated by dividing state 3 by state 4 respiration rates. For more details see "Experimental Procedures." Values are reported as mean ± S.D. For each strain, $n \geq 6$ with mitochondria from each mouse run in triplicate.

Strain	Complex I substrates			Complex II substrates		
	State 4, nmol O_2	State 3, nmol O_2	Respiratory control ratio	State 4, nmol of O_2	State 3, nmol of O_2	Respiratory control ratio
	mg × ml	*mg × ml*		*mg × ml*	*mg × ml*	
ALR	38.34 ± 11.28	138.14 ± 13.09	4.20 ± 1.27	57.86 ± 19.92	200.76 ± 44.71	3.74 ± 0.93
NOD	39.45 ± 11.48	137.64 ± 23.49	4.34 ± 1.11	59.34 ± 16.47	196.05 ± 28.68	3.51 ± 0.96
B6	31.70 ± 8.52	136.54 ± 12.23	4.55 ± 1.30	57.71 ± 19.23	211.28 ± 19.03	3.99 ± 1.33
NOD.mt^{ALR}	38.32 ± 13.02	140.68 ± 23.59	4.31 ± 0.90	60.17 ± 22.98	211.61 ± 31.36	3.82 ± 0.95
ALR.mt^{NOD}	32.47 ± 9.53	136.46 ± 37.31	4.54 ± 0.59	57.16 ± 27.52	200.82 ± 42.58	3.88 ± 0.92

degraded peroxynitrite. Mitochondria were treated with 200 μg/ml glucose and 60 μg/ml or 40 ng/ml glucose oxidase (to generate H_2O_2) in the presence of 0.5 mM sodium nitrite and 100 nM myeloperoxidase and incubated for 1 h. In the presence of hydrogen peroxide, myeloperoxidase has been shown previously to oxidize nitrite to the nitrogen dioxide radical, which is capable of nitrating phenolic protein residues (42). Controls were treated with glucose, sodium nitrite, and myeloperoxidase without glucose oxidase. Complex I enzymatic activity was assayed as described above following incubations with each free radical generator.

Free Radical Treatment for Complex IV Enzymatic Activity Assay—Mitochondria were incubated with 15 μM NO donated by diethylamine NONOate and incubated for 1 h. Complex IV enzymatic activity was assayed as described above following this incubation period.

Statistical Analysis—All values reported are of at least $n = 3$. Significance was determined by one-way analysis of variance using Graphpad Prism 4 for Windows (GraphPad Software, Inc., San Diego, CA). Differences were considered significant at $p < 0.05$.

RESULTS

Single Nucleotide Polymorphism Typing for Conplastic Strains—Pyrosequencing was performed for the A4738C single nucleotide polymorphism to confirm the presence of the *mt-Nd2* allele in both CS. The pyrograms from ALR (supplemental Fig. *A*) and NOD.mt^{ALR} (supplemental Fig. *C*) were equivalent and demonstrated the presence of the *mt-Nd2^a* allele. NOD (supplemental Fig. *B*) and ALR.mt^{NOD} (supplemental Fig. *D*) both contained the *mt-Nd^c* allele. Typing with a panel of microsatellite markers that discriminate ALR from NOD DNA (24) was performed from N_1 to N_7 to confirm the elimination of nuclear DNA contamination in both CS. We did not detect NOD nuclear DNA contamination in the ALR.mt^{NOD} after generation N_6. Likewise, ALR nuclear DNA contamination in the NOD.mt^{ALR} in mice after the N_5 generation was undetectable.

Basal Mitochondrial Enzymatic Activities, Respiration, and Transmembrane Potential—To discern effects of the two *mt-Nd2* alleles on basal mitochondrial function, the mitochondrial ETS was assayed. As shown in Table 1, no differences were measured in enzymatic activities of complexes I–IV when comparing ALR, NOD, B6, NOD.mt^{ALR}, and ALR.mt^{NOD}. Basal mitochondrial respiration supported by complex I or complex II substrates was also assayed (Table 2). State 4 respiration was measured after the addition of mitochondria to the chamber, and state 3 respiration was measured after the addition of ADP. Among the five strains, no differences were detected in either state 4 respiration, state 3 respiration, or the respiratory control ratio (rate of state 3 respiration divided by rate of state 4 respiration) while respiring via complex I or complex II (Table 2). No differences in $\Delta\psi_m$ values supported by complex I or II substrates among the five strains were observed (data not shown).

Mitochondrial ROS Production—An assay that measured changes in fluorescence resulting from oxidation of Amplex Red was used to test the effect of *mt-Nd2^a* on mitochondrial ROS production. ROS production supported by complex I substrates glutamate and malate did not differ among ALR, NOD, B6, and NOD.mt^{ALR}. However, mitochondria isolated from ALR.mt^{NOD} produced ~30% more ROS than each of the other four strains (Fig. 1*A*, *black bars*). Similar results were obtained for ROS production supported by the complex II substrate succinate, and ALR.mt^{NOD} produced ~30% more ROS than the other four strains (Fig. 1*B*, *black bars*).

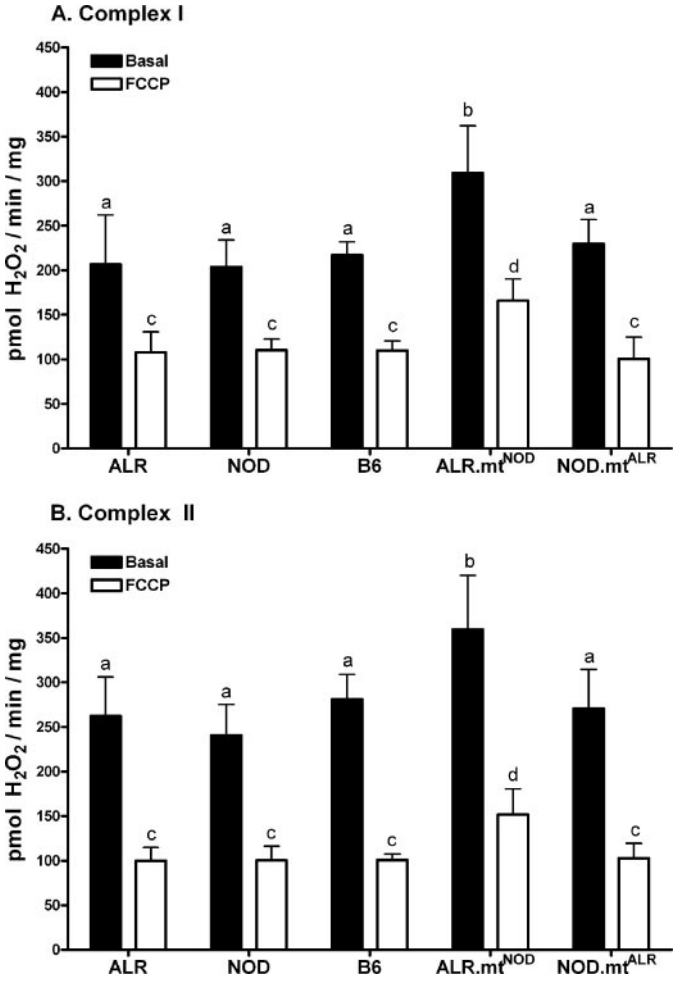


FIGURE 1. **Basal and uncoupled mitochondrial ROS production.** ROS production was detected by following the change in fluorescence because of the oxidation of Amplex Red (see "Experimental Procedures"). Mitochondria were uncoupled using 400 nM FCCP. ROS production was measured for mitochondria respiring on both the complex I substrates glutamate and malate (*A*) the complex II substrate succinate (*B*). Values are reported as means ± S.D. For basal ROS production (*black bars*): ALR, $n = 10$; NOD, $n = 12$; B6, $n = 4$; ALR.mt^{NOD}, $n = 13$; and NOD.mt^{ALR}, $n = 7$. *a versus b*, $p < 0.01$. For uncoupled ROS production (*white bars*), $n = 3$ for all strains. *c versus d*, $p < 0.05$.

To determine whether the observed increase in ROS production in ALR.mt^{NOD} mitochondria was dependent upon the proton motive force (Δp), mitochondrial ROS production was assayed in the presence of 400 nM FCCP. FCCP completely uncouples mitochondrial O_2 consumption from ATP production and dissipates Δp, as it is an efficient proton transporter across the inner mitochondrial membrane. FCCP decreased mitochondrial ROS production from all strains by ~50% when respiring on glutamate and malate (Fig. 1*A*, *white bars*) and 60% on succinate (Fig. 1*B*, *white bars*). However, uncoupled ALR.mt^{NOD} mitochondria respiring on glutamate and malate or on succinate still produced ~30% more ROS than ALR, NOD, NOD.mt^{ALR}, or B6 mitochondria.

Of physiological importance, ROS production was also assayed in the presence of ADP (1.81 mM). Although overall ROS production was decreased by the reduction in $\Delta\psi_m$ due to ATP production, ALR.mt^{NOD} mitochondria still produced 30% more ROS than the other four strains. This increase in ROS

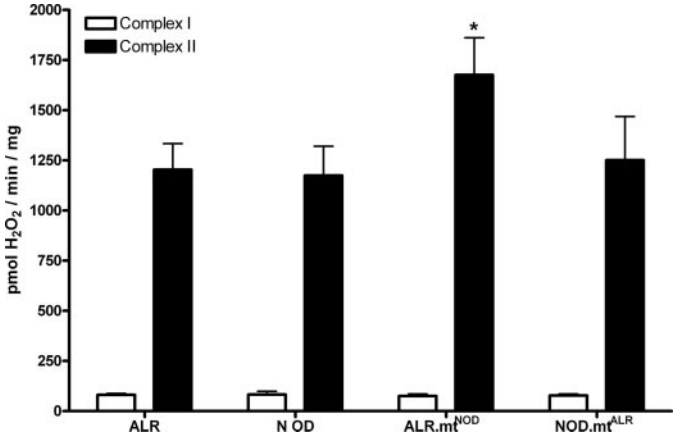


FIGURE 2. **Complex I or complex II supported ROS production for brain mitochondria from ALR, NOD, NOD.mt^{ALR}, and ALR.mt^{NOD}.** Brain mitochondria were isolated as described under "Experimental Procedures," and H_2O_2 production was assayed by monitoring the increase in fluorescence because of the oxidation of Amplex Red. Glutamate and malate were used as the complex I substrates, and succinate was used as the complex II substrate. Values are reported as mean ± S.D. For each strain, $n \geq 3$ with mitochondria from each mouse run in triplicate. *, $p < 0.05$.

production was witnessed when respiration was supported by either complex I or complex II substrates (data not shown).

To determine whether increased H_2O_2 production was present globally in ALR.mt^{NOD}, brain mitochondrial ROS production was assessed. As has been reported previously, when brain mitochondria respire on the complex I substrates glutamate and malate, there is little to no free radical production (43). Therefore, there were no differences in basal complex I substrate-supported H_2O_2 production among ALR, NOD, ALR.mt^{NOD}, and NOD.mt^{ALR} (Fig. 2). However, while respiring on the complex II substrate succinate, ALR.mt^{NOD} mitochondria produced ~30% more H_2O_2 than ALR, NOD, and NOD.mt^{ALR} (Fig. 2).

Effect of NO on Mitochondrial Respiration—To determine the effect of *mt-Nd2^a* on free radical-induced mitochondrial dysfunction, respiring mitochondria were treated with 15 μM NO 30 s after the addition of ADP. Comparisons were based on the percent of control oxygen consumption 1.5 min after the addition of NO. While respiring through complex I or complex II (Table 3), mitochondria from ALR, NOD, B6, ALR.mt-^{NOD}, and NOD.mt^{ALR} were inhibited equally by NO. These results suggest that *mt-Nd2^a* does not alter the susceptibility of respiring mitochondria to NO-induced inhibition.

Effect of Free Radicals on Complex I Enzymatic Activity—To test the theory that the *mt-Nd2^a* allele confers resistance against free radical-induced mitochondrial dysfunction, mitochondria were treated with a variety of different free radicals. Complex I activity was assayed after each treatment. Mitochondria were incubated with either a bolus of NO (donated by diethylamine NONOate) or a steady-state level of NO (donated by DETA-NONOate). Treating mitochondria with 15 μM NO (the concentration used to inhibit respiration) had no effect on complex I activity (data not shown). Increasing the NO concentration to 3 mM and incubation for 2 h did not inhibit complex I activity from any of the mouse strains. Similarly, after a 2-h incubation with a steady-state generation of 1 μM NO, there was no significant decrease in complex I activity (Table 4).

The effects of hydrogen peroxide and superoxide were studied, using xanthine and xanthine oxidase to generate a steady-state level of these ROS. The function of xanthine oxidase was tested by spectrofluorometrically following the oxidation of Amplex Red with an excitation wavelength of 560 nm and an emission wavelength of 590 nm. Using 20 milliunits/ml xanthine oxidase, complex I activity was not significantly decreased from control activity in mitochondria from any of the mouse strains (Table 4). The effect of hydrogen peroxide alone on complex I activity was also studied. No difference in complex I activity was seen when using either 0.25 or 0.50 mM H_2O_2 (data not shown).

Given a previous report suggesting resistance of ALR to protein nitration (38), we also assayed complex I function under conditions known to nitrate phenolic protein residues using varying concentrations of peroxynitrite or heme peroxidase-dependent reactions. As shown in Table 5, 0.25, 0.50, and 1.00 mM peroxynitrite significantly inhibited mitochondrial complex I activity from each mouse strain by ~12, 27.5, and 46%, respectively. Each peroxynitrite concentration produced a significant decrease in complex I activity relative to the other concentrations. However, there was no difference among the complex I activities of the individual mouse strains for any of the peroxynitrite concentrations. Sodium nitrite and myeloperoxidase in the presence of glucose and glucose oxidase (to generate H_2O_2) were used to generate nitrogen dioxide radicals known to nitrate phenolic protein residues (42). The function of glucose oxidase was tested as described above for xanthine oxidase. With 40 ng/ml glucose oxidase, complex I activity was not decreased. With 60 μg/ml glucose oxidase, complex I activity was significantly decreased from control activity by ~24%; however, there was no difference in activity loss among the five mouse strains (Table 4).

TABLE 3

Effect of 15 μM NO on mitochondrial respiration

Respiration was assayed as described under "Experimental Procedures." NO (15 μM) was added 30 s after the addition of 1.81 mM ADP. Values represent percent of O_2 consumption, which was calculated by comparing the concentration of oxygen in the chamber at the time of NO addition to the concentration of oxygen in the chamber 1.5 min after NO addition. Calculations were based on the percent of control oxygen consumption. Values are reported as mean ± S.D. For each strain, $n \geq 3$ with mitochondria from each mouse run in triplicate.

Strain	Complex I	Complex II
ALR	57.52 ± 5.67	57.12 ± 9.70
NOD	57.32 ± 1.55	54.89 ± 8.79
B6	55.50 ± 10.83	54.83 ± 8.27
NOD.mt^{ALR}	55.17 ± 6.12	58.92 ± 4.93
ALR.mt^{NOD}	54.12 ± 5.82	53.87 ± 6.42

Reversibility of NO-induced Respiratory Dysfunction—Given that NO had no effect on the activity of complex I in any of the mouse strains assayed, it was hypothesized that NO reversibly inhibited mitochondrial respiration by acting as a competitive antagonist of molecular oxygen at complex IV. To test whether the NO-induced inhibition was reversible, 15 μM NO was used to inhibit mitochondrial respiration as described above, and following 5 min of inhibition with NO, 1.32 mg/ml Hb was added to bind NO. After the addition of Hb, state 3 respiration resumed and was shown to be not significantly different from the state 3 respiratory rate before the addition of NO (Table 6). Control traces of buffer with and without Hb showed that Hb alone did not alter the oxygen content in the chamber (data not shown). When Hb was added to mitochondria in state 4 respiration, there was no effect on the rate of oxygen consumption (data not shown).

NO, Competitive Antagonism at Complex IV—To confirm that NO was in fact acting as a competitive antagonist of complex IV, mitochondria were incubated with 15 μM NO for 1 h. Complex IV enzymatic activity was then assayed. For mitochondria from each of the strains, NO inhibited complex IV activity by ~95% compared with controls without nitric oxide (Table 7). However, there was no difference in inhibition among the individual strains.

DISCUSSION

The mitochondrial gene NADH dehydrogenase subunit 2 has been associated with resistance to T1D in both humans (6) and mice (23). The protective alleles in both human and mouse differ from the more common allele by a C to A transversion resulting in a leucine to methionine amino acid substitution. In this study, we sought to explore how *mt-Nd2*^{a} alters mitochondrial function to better understand how this allele provides resistance to T1D. We used isolated mitochondria from T1D-resistant ALR mice encoding *mt-Nd2*^{a} and T1D-prone NOD mice with *mt-Nd2*^{c} (cytosine containing NADH dehydrogenase subunit 2 allele) (23). To further investigate the effects of the allelic variation of *mt-Nd2*, we generated two lines of CS as follows: one with ALR nuclear DNA and NOD mtDNA (ALR.mt^{NOD}) and one with NOD nuclear DNA and ALR mtDNA (NOD.mt^{ALR}). B6 mice, encoding *mt-Nd2*^{c}, were used as controls in all experiments.

We first analyzed several parameters of basal mitochondrial function from each of the five strains and only detected a difference when analyzing mitochondrial ROS production. Mitochondria from ALR.mt^{NOD} were shown to produce significantly more ROS than ALR, NOD, B6, and NOD.mt^{ALR}

TABLE 4

Effect of free radicals on mitochondrial complex I enzymatic activity

Complex I activity was assayed as described under "Experimental Procedures." The percent of control complex I activity is displayed after the following: a 2-h incubation with 3 mM NO; a 2-h incubation with a steady-state level of 1 μM NO; a 2-h incubation with 20 milliunits of xanthine oxidase, or a 1-h incubation with 0.5 mM sodium nitrite, 100 nM myeloperoxidase, and 200 μg/ml glucose in the presence of 60 μg/ml or 40 ng/ml glucose oxidase. Values are represented as mean ± S.D. For each strain, $n = 6$ with mitochondria from each mouse run in triplicate.

ROS generator	ALR	NOD	B6	ALR.mt^{NOD}	NOD.mt^{ALR}
NO 3 mM	99.7 ± 10.61	97.0 ± 5.75	100.34 ± 10.44	104.42 ± 10.66	99.24 ± 10.64
Steady-state NO 1 μM	102.84 ± 11.15	100.61 ± 15.56	102.34 ± 18.52	100.92 ± 12.79	97.73 ± 12.49
Xanthine oxidase 20 milliunits/ml	96.87 ± 7.06	97.28 ± 11.30	90.30 ± 14.04	95.55 ± 15.13	96.81 ± 5.32
Glucose oxidase 40 ng/ml	77.61 ± 4.60^a	77.93 ± 10.27^a	77.19 ± 6.88^a	73.19 ± 4.69^a	76.18 ± 8.87^a
Glucose oxidase 60 μg/ml	100.44 ± 4.21	102.43 ± 10.00	96.84 ± 4.20	97.49 ± 8.14	100.23 ± 9.70

$^a p < 0.05$ compared with the 40 ng/ml group.

mitochondria (Fig. 1). The elevated ROS production by liver mitochondria from ALR.mt^{NOD} occurred in both the presence and absence of ADP. ROS production by liver mitochondria after dissipating Δp with FCCP was decreased by ~50% on complex I substrates and 60% on complex II substrates, yet followed a similar trend compared with basal mitochondrial ROS production; uncoupled ALR.mt^{NOD} mitochondria produced significantly more ROS on either complex I (Fig. 1*A*, *white bars*) or complex II (Fig. 1*B*, *white bars*) compared with uncoupled mitochondria from ALR, NOD, B6, or NOD.mt^{ALR}. Mitochondrial transmembrane potential did not differ among the five strains; these results indicate that the increased ROS production in ALR.mt^{NOD} mitochondria is not dependent upon Δp, as the increased ROS production is still present after dissipation of Δp with FCCP.

To determine whether the elevated ROS signal in ALR.mt^{NOD} mitochondria was present globally, we also assayed ROS production in isolated brain mitochondria. As expected, given the well characterized low to absent ROS production using complex I substrates (43, 44), we observed no difference in glutamate- and malate-supported ROS production (Fig. 2). On the contrary, the complex II substrate succinate has been shown to support markedly elevated ROS production from brain mitochondria (43–45). Thus, we found that as in liver, ALR.mt^{NOD} brain mitochondria produced ~30% more ROS supported by succinate (Fig. 2).

Building of the CS allowed us to determine the effects of the *mt-Nd2* alleles in the presence of an alternative nuclear genome. When combined with the NOD nuclear genome, the two alleles of *mt-Nd2* do not result in measurable differences in mitochondrial function. The result of increased ROS production by the ALR.mt^{NOD}, when compared with ALR, NOD, or NOD.mt^{ALR}, demonstrates that nuclear DNA of ALR encodes a factor responsible for the elevated ROS production, and *mt-Nd2*^{a} serves to suppress ROS. The suppression of ROS in ALR compared with ALR.mt^{NOD} may be the result of a structural difference in the ETS, a difference in anti-oxidant defenses potentially resulting from elevated free radical production, or a combination of the two.

It has been suggested that *mt-Nd2*^{a} may confer resistance to T1D by conferring upon mitochondria an increased resistance to dysfunction induced by elevated oxidative stress (6). However, our results did not indicate any protective effect against oxidative stress. NO inhibited respiration to a similar extent in mitochondria from each of the strains (Table 3). Also, *mt-Nd2*^{a} did not confer elevated resistance to complex I enzymatic activity against a bolus of NO, steady-state level of NO, or a steady-state level of hydrogen peroxide and superoxide (Table 4).

We show that peroxynitrite (Table 5) or heme-dependent peroxidase reactions (Table 4) significantly inhibit complex I activity, however, with no significant difference in inhibition among the mitochondria from each different strain. A previous study indicated that ALR mitochondria were not significantly inhibited by peroxynitrite or heme peroxidase-dependent reactions, although mitochondria isolated from ALS were significantly inhibited (38). The discrepancy in the current and previous reports may be due to the use of cardiac mitochondria in the previous study or the low antioxidant levels of the ALS mouse. Taken together, our results indicate that mitochondria encoding *mt-Nd2*^{a} do not possess elevated resistance to oxidative stress, because they were equally susceptible to free radical-mediated inhibition of respiration or complex I activity as mitochondria encoding *mt-Nd2*^{c}.

Riobo *et al.* (46) have reported that the NO-mediated respiratory inhibition was not fully reversible with mitochondria respiring on complex I substrates because of peroxynitrite formation and subsequent nitration of complex I, whereas NO-mediated respiratory inhibition of mitochondria respiring on complex II substrates was completely reversible. After inhibition of mitochondria with NO, we added exogenous Hb to bind NO. We found that with mitochondria respiring on either complex I and complex II substrates (Table 6), NO inhibition of mitochondrial respiration was completely reversible, with state 3 respiration rates before the addition of NO not differing significantly from rates after the binding of NO by Hb. However, when incubating mitochondria with 15 μM NO, complex IV enzymatic activity was inhibited ~95%, with the inhibition not differing significantly among the five strains (Table 7). These results suggested that significant amounts of peroxynitrite were not forming under our reaction conditions, and NO was acting

TABLE 5

Effect of peroxynitrite on complex I activity

Mitochondria (200 μg/ml) were treated with 0.25, 0.50, or 1.00 mM peroxynitrite for 1 h. Complex I enzymatic activity was assayed as described under "Experimental Procedures." Values represent percent of control complex I activity and are reported as mean ± S.D. For each strain $n \geq 3$ with mitochondria from each mouse run in triplicate. [a, b, c], $p \leq 0.05$.

Strain	0.25 mM	0.50 mM	1.00 mM
ALR	87.81 ± 5.68[a]	73.76 ± 6.36[b]	55.27 ± 5.19[c]
NOD	90.49 ± 5.49[a]	72.10 ± 6.56[b]	52.67 ± 8.06[c]
B6	88.01 ± 5.90[a]	73.52 ± 5.62[b]	52.23 ± 6.45[c]
NOD.mt^{ALR}	89.46 ± 4.41[a]	72.27 ± 5.34[b]	55.77 ± 4.65[c]
ALR.mt^{NOD}	85.30 ± 4.71[a]	70.91 ± 3.77[b]	54.75 ± 6.31[c]

TABLE 6

Reversibility of NO induced respiratory dysfunction

Oxygen consumption was measured using a Clark oxygen-type electrode, and state 3 was induced by the addition of 1.81 mM ADP (see "Experimental Procedures"). NO (15 μM) was added 30 s after the addition of ADP to inhibit mitochondrial respiration. After 5 min of incubation with NO, Hb (1.32 mg/ml) was added to bind NO. Percent of control values represent the comparison of the state 3 respiration rate after the addition of Hb to the initial state three respiration rate before the addition of NO. Values are reported as means ± S.D. For each strain, $n = 3$ with mitochondria from each strain run in triplicate.

Strain	Complex I substrates		Complex II substrates	
	State 3 (pre-NO)	State 3 (post-NO)	State 3 (pre-NO)	State 3 (post-NO)
ALR	121.65 ± 20.72	113.50 ± 26.24	188.48 ± 26.70	177.33 ± 32.53
NOD	119.69 ± 7.16	115.76 ± 10.23	184.44 ± 18.59	169.70 ± 20.80
B6	131.02 ± 9.99	122.74 ± 14.38	176.98 ± 14.87	165.56 ± 7.78
NOD.mt^{ALR}	131.78 ± 11.09	123.92 ± 10.37	194.94 ± 9.89	188.01 ± 3.70
ALR.mt^{NOD}	129.80 ± 15.97	122.89 ± 9.82	201.72 ± 13.84	188.60 ± 7.16

TABLE 7

Effect of 15 μM NO on complex IV activity

Mitochondria (2.5 μg) were treated with 15 μM NO and incubated for 1 h. Complex IV enzymatic activity was assayed as described under "Experimental Procedures." Values are reported as mean ± S.D. For each strain, $n \geq 3$ with mitochondria from each mouse run in triplicate.

Strain	Percent of control complex IV activity
ALR	95.18 ± 2.09
NOD	94.08 ± 1.70
B6	95.18 ± 2.07
NOD.mt^{ALR}	93.46 ± 1.85
ALR.mt^{NOD}	94.76 ± 1.03

as a competitive antagonist with molecular oxygen to inhibit the mitochondrial respiration, as demonstrated previously (47).

In contrast to the proposed mechanism of elevated resistance to oxidative stress by Uchigata *et al.* (6), we suggest that allelic variants of *mt-Nd2* play a critical role in the production of ROS by the mitochondrial ETS. Our data show that ALR.mt^{NOD} mitochondria produce ~30% more ROS than both parental strains or the reciprocal CS, indicating that an interaction is occurring between a protein encoded by the nuclear DNA of ALR and the protein product of *mt-Nd2^c*. Furthermore, based on elevated ROS production supported by either complex I or complex II substrates (Fig. 1), it is likely that an altered interaction between complexes I and III results in increased ROS. It is unlikely that the increased complex II-supported ROS production in ALR.mt^{NOD} mitochondria is solely due ROS production from complex I redox centers, because the increase is present after the addition of the uncoupler FCCP, which prevents reverse electron flow back to complex I (Fig. 1).

This mechanism may also account for the results obtained in the initial backcross study implicating the combination of *mt-Nd2^a* and ALR nuclear genes in resistance against T1D (23). In that study, reciprocal backcross mice were generated: ((NOD/LtDvs (♀) × ALR/Lt (♂))F1 (♀) × NOD/LtDvs (♂)) and ((ALR/Lt (♀) × NOD/LtDvs (♂))F1 (♀) × NOD/LtDvs (♂)). The former backcross population encoded *mt-Nd2^c* and developed spontaneous T1D at a 4-fold higher rate than the latter population encoding *mt-Nd2^a* (23). *mt-Nd2^c* combined with ALR nuclear DNA on chromosome 3, 8, or 17 in this backcross may have resulted in an elevated ROS production leading to increased free radical defenses. Support for this hypothesis has been provided by previous studies that have associated very low β cell antioxidant defenses with increased susceptibility and increased islet antioxidants with resistance against NOD-derived immune effectors (27, 32, 48–51, 53–57).

A potential candidate gene is *Ndufb7*, which is contained within the *Idd22* confidence interval on chromosome 8 previously reported to contribute to the resistance of ALR to T1D (24). The phosphorylation of this complex I subunit by pyruvate dehydrogenase kinase has been reported to lead to an increased mitochondrial superoxide production (58). Likewise, the suppressor of superoxide production locus (*Susp*, encoded on chromosome 3), which has been shown to account for resistance against T1D in ALR, contains electron transfer flavoprotein dehydrogenase (*Etfdh*), a protein that resides in the inner mitochondrial membrane. Electrons can be transferred by ETFDH to ubiquinone entering the mitochondrial respiratory chain (59). Deficiency in ETFDH leads to accumulation of glutaric acid resulting in increased oxidative stress (52). We have measured a decrease in *Etfdh* expression in ALR compared with NOD (data not shown). Sequence variation of *Etfdh* in ALR leading to reduced levels of ETFDH in the mitochondrial inner membrane could lead to increased mitochondrial ROS production in combination with *mt-Nd2^c* that is eliminated when combined with *mt-ND2^a*.

In summary, the two alleles of *mt-Nd2* studied here do not alter basal mitochondrial function, and only exhibit differences in mitochondrial ROS production when *mt-Nd2^c* is combined with specific ALR-derived nuclear factors. As measures of mitochondria function were equal after incubation with different radical species when comparing the five strains, it is unlikely that *mt-Nd2^a* itself scavenges ROS. Rather, these findings suggest that an elevation in ROS production resulting from a structural change in the ETS when *mt-Nd2^c* is combined with ALR nuclear DNA is responsible for a decrease in the incidence of T1D. Further, the presence of *mt-Nd2^a* results in a lower ROS production and subsequent protection from the occurrence of T1D.

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REFERENCES

1. Yao, Y. G., Kong, Q. P., and Zhang, Y. P. (2002) *Hum. Genet.* **111,** 462–463
2. Tanaka, M., Gong, J. S., Zhang, J., Yoneda, M., and Yagi, K. (1998) *Lancet* **351,** 185–186
3. Matsunaga, H., Tanaka, Y., Tanaka, M., Gong, J. S., Zhang, J., Nomiyama, T., Ogawa, O., Ogihara, T., Yamada, Y., Yagi, K., and Kawamori, R. (2001) *Diabetes Care* **24,** 500–503
4. Kokaze, A., Ishikawa, M., Matsunaga, N., Yoshida, M., Sekine, Y., Sekiguchi, K., Harada, M., Satoh, M., Teruya, K., Takeda, N., Fukazawa, S., Uchida, Y., and Takashima, Y. (2004) *J. Hum. Hypertens.* **18,** 41–45
5. Takagi, K., Yamada, Y., Gong, J. S., Sone, T., Yokota, M., and Tanaka, M. (2004) *Atherosclerosis* **175,** 281–286
6. Uchigata, Y., Okada, T., Gong, J. S., Yamada, Y., Iwamoto, Y., and Tanaka, M. (2002) *Diabetes Care* **25,** 2106
7. Silva, J. P., Kohler, M., Graff, C., Oldfors, A., Magnuson, M. A., Berggren, P. O., and Larsson, N. G. (2000) *Nat. Genet.* **26,** 336–340
8. Wollheim, C. B., and Maechler, P. (2002) *Diabetes* **51,** Suppl. 1, 37–42
9. Barnhart, B. C., Alappat, E. C., and Peter, M. E. (2003) *Semin. Immunol.* **15,** 185–193
10. Thorburn, A. (2004) *Cell. Signal.* **16,** 139–144
11. Silva, D. G., Petrovsky, N., Socha, L., Slattery, R., Gatenby, P., and Charlton, B. (2003) *J. Immunol.* **170,** 4996–5002
12. Silva, D. G., Socha, L., Charlton, B., Cowden, W., and Petrovsky, N. (2003) *Ann. N. Y. Acad. Sci.* **1005,** 161–165
13. Petrovsky, N., Silva, D., Socha, L., Slattery, R., and Charlton, B. (2002) *Ann. N. Y. Acad. Sci.* **958,** 204–208
14. Chervonsky, A. V., Wang, Y., Wong, F. S., Visintin, I., Flavell, R. A., Janeway, C. A., and Matis, L. (1997) *Cell* **89,** 17–24
15. Suarez-Pinzon, W. L., Power, R. F., and Rabinovitch, A. (2000) *Diabetologia* **43,** 1149–1156
16. Itoh, N., Imagawa, A., Hanafusa, T., Waguri, M., Yamamoto, K., Iwahashi, H., Moriwaki, M., Nakajima, H., Miyagawa, J., Namba, M., Makino, S., Nagata, S., Kono, N., and Matsuzawa, Y. (1997) *J. Exp. Med.* **186,** 613–618
17. Friedrich, T., Abelmann, A., Brors B., Guenebaut, V., Kintscher, L., Leonard, K., Rasmussen, T., Scheide, D., Schlitt, A., Schulte, U., and Weiss, H. (1998) *Biochem. Biophys. Acta* **1365,** 215–219
18. Friedrich, T., Steinmüller, K., and Weiss, H. (1995) *FEBS Lett.* **367,** 107–111
19. Flemming, D., Stolpe, S., Schneider, D., Hellwig, P., and Friedrich, T.

(2005) *J. Mol. Microbiol. Biotechnol.* **10,** 208–222
20. Mathiesen, C., and Hägerhäll, C. (2002) *Biochem. Biophys. Acta* **1556,** 121–132
21. Friedrich, T. (2001) *J. Bioenerg. Biomembr.* **33,** 169–177
22. Gabaldon, T. R. D., and Huynen, M. (2005) *J. Mol. Biol.* **348,** 857–870
23. Mathews, C. E., Leiter, E. H., Spirina, O., Bykhovskaya, Y., Gusdon, A. M., Ringquist, S., and Fischel-Ghodsian, N. (2005) *Diabetologia* **48,** 261–267
24. Mathews, C. E., Graser, R. T., Bagley, R. J., Caldwell, J. W., Li, R., Churchill, G. A., Serreze, D. V., and Leiter, E. H. (2003) *Immunogenetics* **55,** 491–496
25. Sekiguchi, F., Ishibashi, K., Katoh, H., Kawamoto, Y., and Ino, T. (1990) *Exp. Anim.* (*Tokyo*) **39,** 269–272
26. Mathews, C. E., and Leiter, E. H. (1999) *Free Radic. Biol. Med.* **27,** 449–455
27. Mathews, C. E., Graser, R. T., Savinov, A. Y., Serreze, D. V., and Leiter, E. H. (2001) *Proc. Natl. Acad. Sci., U. S. A.* **98,** 235–240
28. Buttke, T., and Sandstrom, P. A. (1994) *Immunol. Today* **15,** 7–10
29. Briehl, M. M., Cotgreave, I. A., and Powis, G. (1995) *Cell Death Differ.* **2,** 41–46
30. Briehl, M. M., and Baker, A. F. (1996) *Cell Death Differ.* **3,** 63–70
31. Mathews, C. E., and Leiter, E. H. (1999) *Diabetes* **48,** 2189–2196
32. Mathews, C. E., Suarez-Pinzon, W. L., Baust, J. J., Strynadka, K., Leiter, E. H., and Rabinovitch, A. (2005) *J. Immunol.* **175,** 1248–1256
33. Ho, E., Chen, G. M., and Bray, T. M. (1999) *FASEB J.* **13,** 1845–1854
34. Lizard, G., Gueldry, S., Sordet, O., Monier, S., Athias, A., Miguet, C., Bessede, G., Lemaire, S., Solary, E., and Gambert, P. (1998) *FASEB J.* **12,** 1654–1663
35. Saitoh, M., Nishitoh, H., Fujii, M., Takeda, K., Tobiume, K., Sawada, Y., Kawabata, M., Miyazono, K., and Ichijo, H. (1998) *EMBO J.* **17,** 2596–2606
36. Genova, M. L., Pich, M. M., Biondi, A., Bernacchia, A., Falasca, A., Bovina, C., Formiggini, G., Castelli, G. P., and Lenaz, G. (2003) *Exp. Biol. Med.* (*Maywood*) **228,** 506–513
37. Nomura, K., Imai, H., Koumura, T., and Nakagawa, Y. (2001) *Biol. Signals Recept.* **10,** 81–92
38. Turko, I. V., Li, L., Aulak, K. S., Stuehr, D. J., Chang, J. Y., and Murad, F. (2003) *J. Biol. Chem.* **278,** 33972–33977
39. Pomerleau, D. P., Bagley, R. J., Serreze, D. V., Mathews, C. E., and Leiter, E. H. (2005) *Diabetes* **54,** 1603–1606
40. Birch-Machin, M., and Turnbull, D. M. (2001) in *Mitochondria* (Pon, L., and Schon, E. A., eds) Academic Press, San Diego
41. Votyakova, T. V., and Reynolds, I. J. (2005) *J. Neurochem.* **93,** 526–537
42. van der Vliet, A., Eiserich, J. P., Halliwell, B., and Cross, C. E. (1997) *J. Biol. Chem.* **272,** 7617–7625
43. Votyakova, T. V., and Reynolds, I. J. (2001) *J. Neurochem.* **79,** 266–277
44. Kudin, A. P., Bimpong-Buta, N. Y.-B., Vielhaber, S., Elger, C. E., and Kunz, W. S. (2003) *J. Biol. Chem.* **279,** 4127–4135
45. Kwong, L. K., and Sohal, R. S. (1998) *Arch. Biochem. Biophys.* **350,** 118–126
46. Riobo, N. A., Clementi, E., Melani, M., Boveris, A., Cadenas, E., Moncada, S., and Poderoso, J. J. (2001) *Biochem. J.* **359,** 139–145
47. Jekabsone, A., Ivanoviene, L., Brown, G. C., and Borutaite, V. (2003) *J. Mol. Cell. Cardiol.* **35,** 803–809
48. Lenzen, S., Drinkgern, J., and Tiedge, M. (1996) *Free Radic. Biol. Med.* **20,** 463–466
49. Tiedge, M., Lortz, S., Drinkgeer, J., and Lenzen, S. (1997) *Diabetes* **46,** 1733–1742
50. Gale, E. A. (1996) *Horm. Res.*(*Basel*) **45,** Suppl. 1, 39–43
51. Horio, F., Fukuda, M., Katoh, H., Petruzzelli, M., Yano, N., Rittershaus, C., Bonnerweir, S., and Hattori, M. (1994) *Diabetologia* **37,** 22–31
52. de Oliveira Marques, F., Hagen, M. E., Pederzolli, C. D., Sgaravatti, A. M., Durigon, K., Testa, C. G., Wannmacher, C. M., de Souza Wyse, A. T., Wajner, M., and Dutra-Filho, C. S. (2003) *Brain Res.* **964,** 153–158
53. Nerup, J., Mandrup-Poulsen, T., Molvig, J., Helquist, S., Wogensen, L., and Egeberg, J. (1988) *Diabetes Care* **11,** 16–23
54. Suarez-Pinzon, W. L., Mabley, J. G., Strynadka, K., Power, R. F., Szabo, C., and Rabinovitch, A. (2001) *J. Autoimmun.* **16,** 449–455
55. Tabatabaie, T., Waldon, A. M., Jacob, J. M., Floyd, R. A., and Kotake, Y. (2000) *Biochem. Biophys. Res. Commun.* **273,** 699–704
56. Bertera, S., Crawford, M. L., Alexander, A. M., Papworth, G. D., Watkins, S. C., Robbins, P. D., and Trucco, M. (2003) *Diabetes* **52,** 387–393
57. Hotta, M., Tashiro, F., Ikegami, H., Niwa, H., Ogihara, T., Yodoi, J., and Miyazaki, J. (1998) *J. Exp. Med.* **188,** 1445–1451
58. Raha, S., Myint, T., Johnstone, L., and Robinson, B. H. (2002) *Free Radic. Biol. Med.* **32,** 421–430
59. Zhang, J., Frerman, F. E., and Kim, J. J. (2006) *Proc. Natl. Acad. Sci. U. S. A.* **103,** 16212–16217
60. Ringquist, S., Alexander, A. M., Styche, A., Pecoraro, C., Rudert, W. A., and Trucco, M. (2004) *Hum. Immunol.* **65,** 163–174